2025 IEEE International Conference on Communications Workshops (ICC Workshops) | 979-8-3315-9624-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCWorkshops67674.2025.11162345

Impact of Synchronization Offsets and CSI Feedback Delay in Distributed MIMO Systems

Kumar Sai Bondada*, Daniel J. Jakubisin*†, R. Michael Buehrer*
*Wireless@VT, Bradley Department of ECE, Virginia Tech, Blacksburg, VA, USA
†Virginia Tech National Security Institute, Blacksburg, VA, USA

Abstract—**The main challenges of distributed MIMO systems lie in achieving highly accurate synchronization and ensuring the availability of accurate channel state information (CSI) at distributed nodes. This paper analytically examines the effects of synchronization offsets and CSI feedback delays on system capacity, providing insights into how these affect the coherent joint transmission gain. The capacity expressions are first derived under ideal conditions, and the effects of synchronization offsets and feedback delays are subsequently incorporated. This analysis can be applied to any distributed MIMO architecture. A comprehensive study, including system models and simulations evaluating the analytical expressions, is presented to quantify the capacity degradation caused by these factors. This study provides valuable insights into the design and performance of distributed MIMO systems. The analysis shows that time and frequency offsets, along with CSI feedback delay, cause inter-layer interference. Additionally, time offsets result in inter-symbol interference.**

Index Terms—**6G, Distributed MIMO, Distributed Antenna System, Capacity, Synchronization, Channel State Information feedback delay, Stationary process**

I. Introduction & Related Works

Sixth-generation (6G) wireless networks are envisioned to deliver immersive, hyper-reliable, and low-latency communication, alongside massive and ubiquitous connectivity to meet the demands of emerging applications. Compared to fifth-generation (5G) networks, 6G is anticipated to achieve a 5- to 10-fold improvement in spectral efficiency and peak data rates reaching 1 Tbps [1]–[3]. This significant enhancement is motivated by novel use cases, such as autonomous vehicles, intelligent transportation systems, industrial automation, environmental sensing, and augmented/virtual reality (AR/VR). These advancements necessitate a paradigm shift in communication networks toward distributed and collaborative architectures capable of supporting the stringent performance requirements of 6G.

Distributed MIMO systems are emerging as one of the key enablers for 6G communications [4], [5]. Prior research has demonstrated the potential of these systems to achieve significant improvements in both spectral efficiency and energy consumption [6]. Various distributed MIMO architectures [7]–[9] have been proposed under different terminologies, each focusing on objectives such as improving spectral and energy efficiency [10], mitigating inter-cell interference [11]–[14], ensuring uniform quality of service across the network [15]–[17], extending communication range [18], realizing the theoretical gains of massive MIMO by addressing form factor constraints, and offering macro-diversity. Theoretical bounds on capacity by scaling up the number of distributed nodes have been studied in [19]. The evolution of distributed MIMO systems has transitioned from connecting distributed mobile radios to a central processing unit via high-speed wired connections to exploring wireless connections enabled by the large bandwidths available in millimeter-wave frequencies [20]–[22].

Although distributed MIMO systems promise significant improvements in spectral and energy efficiency, they face several challenges, such as achieving highly accurate synchronization and ensuring the availability of accurate channel state information (CSI) at distributed radios (nodes) for coherent joint transmission [8], [9], [23]. Most prior works assume perfect synchronization or perfect CSI availability at the nodes when analyzing the proposed distributed MIMO systems. However, only a limited number of studies [24]–[26] have explored the impact of synchronization errors, often considering only timing errors and evaluating their effects through bit error rate (BER) analysis. The work in [27] examined the effects of initial phase offset errors in a multiple-transmitter scenario with a single receiver, where each transmitter and the receiver are equipped with a single antenna. While these studies provide valuable insights, they primarily address limited scenarios or focus on specific types of errors, leaving a gap in understanding how synchronization errors and CSI feedback delays jointly impact the capacity of distributed MIMO systems, especially in multi-antenna configurations. The current paper studies the capacity performances of coherent joint transmission by analyzing the impact of synchronization errors and CSI feedback delays on the capacity of distributed MIMO systems with multiple antenna transceivers. Specifically, it quantifies the degradation in capacity caused by these factors and proposes a detailed analysis framework to better understand their implications for distributed MIMO performance. The proposed analysis is applicable to any distributed MIMO architecture.

The remainder of this paper is organized as follows: In Section II, we describe the system model under ideal conditions and analyze the impact of synchronization errors and CSI feedback delay. Section III presents the capacity expressions considering these effects. Section IV provides simulations numerically evaluating the analytical expressions and insights derived from the analysis. Finally, Section V concludes the paper with key findings and discusses potential future research directions.

Notation: Boldface uppercase letters denote matrices; boldface lowercase letters denote vectors.

II. System Model and Architecture

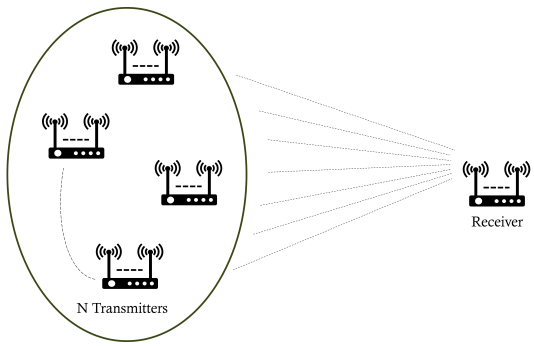


Fig. 1: A group of N cooperating transmitters coherently joint transmitting to the receiver.

We consider a system consisting of multiple N transmitters cooperating in coherent joint transmission to the receiver as shown in Fig. 1. All the transmitters have the same transmit power level, and each transmitter and receiver has the same number of antennas. We assume an ideal scenario, where the transmit data and precoder information are available at all the transmitters to focus on the impact of synchronization offsets and CSI feedback delays. For an analysis of capacity accounting for wireless information sharing among the transmitters, see our prior work [5]. The received signal at time t can be represented as

$$\mathbf{y}(t) = \sum_{i=1}^{N} \mathbf{H}_i(t)\mathbf{x}_i(t) + \mathbf{n}(t), \tag{1}$$

where $\mathbf{H}_i(t)$ represents the channel matrix between the i-th transmitter and the receiver whose elements follow uncorrelated complex Gaussian distribution with zero mean and unit variance, $\mathbf{x}_i(t) = \mathbf{F}_i(t)\mathbf{s}(t)$ is the i-th transmit signal vector with $\mathbf{F}_i(t)$ being the individual precoder of the i-th transmitter, $\mathbf{s}(t)$ is the common transmitted data signal vector and $\mathbf{n}(t)$ is the additive white Gaussian noise (AWGN) signal vector ($\mathbf{n}(t) \sim \mathcal{CN}(0, \mathbf{I}_{N_r})$). Substituting $\mathbf{x}_i(t)$ into the Eq. 1, we have

$$\mathbf{y}(t) = \sum_{i=1}^{N} \mathbf{H}_i(t)\mathbf{F}_i(t)\mathbf{s}(t) + \mathbf{n}(t).$$

This can also be represented in matrix form as

$$\mathbf{y}(t) = \begin{bmatrix}\mathbf{H}_1(t) & \mathbf{H}_2(t) & \cdots & \mathbf{H}_N(t)\end{bmatrix} \begin{bmatrix}\mathbf{F}_1(t) \\ \mathbf{F}_2(t) \\ \vdots \\ \mathbf{F}_N(t)\end{bmatrix} \mathbf{s}(t) + \mathbf{n}(t).$$

Simplifying further, we have

$$\mathbf{y}(t) = \mathbf{H}(t)\mathbf{F}(t)\mathbf{s}(t) + \mathbf{n}(t), \tag{2}$$

where

$$\mathbf{H}(t) = \begin{bmatrix}\mathbf{H}_1(t) & \mathbf{H}_2(t) & \cdots & \mathbf{H}_N(t)\end{bmatrix},$$

and

$$\mathbf{F}(t) = \begin{bmatrix}\mathbf{F}_1(t)^{\mathrm{T}} & \mathbf{F}_2(t)^{\mathrm{T}} & \cdots & \mathbf{F}_N(t)^{\mathrm{T}}\end{bmatrix}^{\mathrm{T}}.$$

The N_t, N_r are considered as the number of transmit and receive antennas of the transceivers, respectively. The dimensions of $\mathbf{H}(t), \mathbf{F}(t)$, and $\mathbf{s}(t)$ are $N_r \times N \cdot N_t$, $N \cdot N_t \times N_s$, and $N_s \times 1$, respectively, where N_s is the maximum number of independent data streams that can be transmitted, given by $N_s = \min(N \cdot N_t, N_r)$.

A. Ideal Scenario and Challenges

In an ideal scenario, all transmitters are assumed to transmit at a specified time with identical carrier frequencies (synchronized oscillators). The precoder, $\mathbf{F}(t)$, is designed to maximize the received signal power at the receiver. However, in practical systems, imperfections in the oscillators at the transmitters result in misalignment of time references and carrier frequencies [28]. The precoder design typically involves transmitting pilot signals either from the receiver to all transmitters or from all transmitters to the receiver. These pilots are used to estimate the channel, based on which the precoder is computed. Nevertheless, this process introduces a delay in the dissemination of the precoder information to all transmitters. Additionally, due to the dynamic nature of wireless environments, the channel conditions at the time t of coherent joint transmission differ from those observed during the pilot transmission at time $(t - \tau)$, where τ represents the feedback delay. These inconsistencies will significantly affect system performance, reducing received signal strength and causing inter-layer and inter-symbol interference (ISI).

B. Impact of channel state information feedback delay

Assuming perfect synchronization, the precoder is computed based on the channel at time $t-\tau$. Consequently, the received signal at the receiver is expressed as

$$\mathbf{y}(t) = \mathbf{H}(t)\mathbf{F}(t-\tau)\mathbf{s}(t) + \mathbf{n}(t),$$

where $\mathbf{F}(t-\tau)$ is the delayed precoder. The elements of the channel $\mathbf{H}(t)$ exhibit the properties of a strict-sense stationary process, which satisfies the following

$$\mathbb{E}[\mathbf{H}_{i,j}(t)] = 0, \quad \mathbb{E}[\mathbf{H}_{i,j}^{\mathrm{H}}(t)\mathbf{H}_{i,j}(t)] = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{otherwise.} \end{cases},$$

$$\rho_{i,j}(\tau) = \mathbb{E}[\mathbf{H}_{i,j}^{\mathrm{H}}(t)\mathbf{H}_{i,j}(t-\tau)] = \begin{cases} J_0(2\pi f_d \tau), & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

where J_0 is the zeroth-order Bessel function of the first kind, and f_d is the Doppler frequency shift. The channel $\mathbf{H}(t)$ can be decomposed using the auto-regressive process [29]–[31] as

$$\mathbf{H}(t) = \mathbf{E}(t) + \rho(\tau)\mathbf{H}(t-\tau),$$

where $\mathbf{E}(t)$ represents the residual complex Gaussian distributed matrix random process with its elements following

$$\mathbf{E}_{i,j}(t) \sim \mathcal{CN}(0, 1-\rho^2(\tau))$$

and is independent of the channel $\mathbf{H}(t-\tau)$. Substituting this into the signal model given in Eq. 2, we obtain

$$\mathbf{y}(t) = \mathbf{E}(t)\mathbf{F}(t-\tau)\mathbf{s}(t) + \rho(\tau)\mathbf{H}(t-\tau)\mathbf{F}(t-\tau)\mathbf{s}(t) + \mathbf{n}(t).$$

Here, the term $\mathbf{E}(t)\mathbf{F}(t-\tau)\mathbf{s}(t)$ represents the interference component causing inter-layer interference since $\mathbf{E}(t)$ is unknown, and the precoder built from $\mathbf{H}(t-\tau)$ is independent of $\mathbf{E}(t)$, whereas $\rho(\tau)\mathbf{H}(t-\tau)\mathbf{F}(t-\tau)\mathbf{s}(t)$ constitutes the signal component.

C. Impact of synchronization errors

The analysis assumes that synchronization among the transmitters has been performed, but residual time-frequency offsets are still present. The synchronization procedure involves one transmitter acting as the reference, to which all other transmitters synchronize in both time and frequency. Considering the impact of time and frequency synchronization errors, the received signal model is further modified to include independent time and frequency offsets

$$\mathbf{y}(t) = \sum_{i=1}^{N} \mathbf{H}_i(t)\mathbf{F}_i(t-\tau-T_i)\mathbf{s}(t-T_i)e^{j2\pi f_i(t-T_i)} + \mathbf{n}(t),$$

where T_i is the time offset and f_i is the frequency offset at the i-th transmitter. The frequency offset has two phase terms, one is the initial phase offset $-2\pi f_i T_i$, and the second is the phase rotation over time $2\pi f_i t$. Assuming the phase rotations caused by frequency offsets f_i are minimal during the coherent joint transmission, we have

$$\mathbf{y}(t) = \sum_{i=1}^{N} \mathbf{H}_i(t)\mathbf{F}_i(t-\tau)\mathbf{s}(t-T_i)e^{j\phi_i} + \mathbf{n}(t),$$

where ϕ_i is the initial random phase offset associated with i-th transmitter frequency offset f_i. Sampling the received signal with an optimum sampling interval T_s, we obtain the discretized version of the received signal as

$$\mathbf{y}(k) = \sum_{i=1}^{N} \mathbf{H}_i(k)\mathbf{F}_i(\bar{k})\mathbf{s}(k-\eta_i)e^{j2\pi f_i k T_s} + \mathbf{n}(k), \quad (3)$$

where $\eta_i = \frac{T_i}{T_s}$ is the normalized time offset and $\bar{k} = k - T$, where $T = \frac{\tau}{T_s}$ indicates delay in samples. Using linear interpolation, the transmitted signal $\mathbf{s}[k-\eta_i]$ can be expressed as

$$\mathbf{s}(k-\eta_i) = (1-\eta_i)\mathbf{s}(k) + \eta_i\mathbf{s}(\tilde{k}),$$

$$\text{where } \tilde{k} = \begin{cases} k+1, & \text{if } \eta_i < 0, \\ k-1, & \text{if } \eta_i > 0. \end{cases}$$

Here, $\eta_i < 0$ indicates that data transmission from the corresponding transmitter started early, while $\eta_i > 0$ indicates that the data transmission was delayed with respect to a transmitter that acts as a reference.

III. Capacity Analysis

The spectral efficiency is expressed as

$$R = \log_2 \det(\mathbf{I} + \text{SINR}),$$

where SINR is the Signal-to-Interference-plus-Noise Ratio, where

$$\text{SINR} = \frac{\mathbf{S}}{\mathbf{I}_{\text{int}} + \mathbf{N}}$$

is measuring the quality of the received signal $\mathbf{S}$ (Signal power) in the presence of both interference power $\mathbf{I}_{\text{int}}$ and noise variance $\mathbf{N}$. The interference arises from the non-coherent combining of the transmitted signals from multiple transmitters due to the synchronization offsets and CSI feedback delay.

Expanding $\mathbf{s}(k-\eta_i)$ in Eq. 3, we have the received signal

$$= \sum_{i=1}^{N} \mathbf{H}_i(k)\mathbf{F}_i(\bar{k}) \left[(1-\eta_i)\mathbf{s}(k) + \eta_i\mathbf{s}(\bar{k})\right] e^{j\phi_i} + \mathbf{n}(k)$$

$$= \sum_{i=1}^{N} \underbrace{\mathbf{H}_i(k)\mathbf{F}_i(\bar{k})(1-\eta_i)\mathbf{s}(k)e^{j\phi_i}}_{\text{Signal component with Inter-layer Interference}} + \sum_{i=1}^{N} \underbrace{\mathbf{H}_i(k)\mathbf{F}_i(\bar{k})\eta_i\mathbf{s}(\tilde{k})e^{j\phi_i}}_{\text{ISI component}} + \mathbf{n}(k)$$

Focusing on the signal component, moving the phase $e^{j\phi_i}$ and time offsets η_i into the summation, and extracting $\mathbf{s}(k)$ as a common factor, we have the Signal part as

$$\sum_{i=1}^{N} \mathbf{H}_i(k)\mathbf{F}_i(\bar{k})(1-\eta_i)\mathbf{s}(k)e^{j\phi_i} = \sum_{i=1}^{N} \mathbf{H}_i(k)\theta_i(k)\phi_i(k)\mathbf{F}_i(\bar{k})\mathbf{s}(k),$$

where $\phi_i(k)$ and $\theta_i(k)$ represents the diagonal matrices of size N_T with phase offsets and time offsets respectively for the i-th transmitter. The phase and time offsets among the antennas of the i-th transmitter are equal because their RF chains are driven by a common oscillator. Finally, the received signal model with interference can be expressed as

$$\mathbf{y}(k) = \underbrace{\mathbf{H}(k)\theta(k)\phi(k)\mathbf{F}(\bar{k})\mathbf{s}(k)}_{\text{Signal component with Inter-layer Interference}} + \underbrace{\mathbf{H}(k)\psi(k)\phi(k)\mathbf{F}(\bar{k})\mathbf{s}(\tilde{k})}_{\text{ISI component}} + \mathbf{n}(k).$$

The phase term $\phi(k)$ is a diagonal matrix containing the phase offsets from all transmitter antennas, where each transmitter has identical phase offsets. The time offset term $\theta(k)$ is a diagonal matrix containing $1-\eta_i$, while the other time offset term $\psi(k)$ contains the corresponding time offsets η_i. Because of the $\phi(k)$, $\theta(k)$, and delay in $\bar{k}$, the precoder cannot effectively align with the channel in the Signal component. As a result, the signals from independent data streams bleed into one another, causing inter-layer interference. We consider the SVD-based precoding where the $\mathbf{F}(\bar{k})$ precoding matrix is designed using the first N_s right singular vectors corresponding to the positive singular values. Because of the time offset term $\psi(k)$, the symbols either from time index $k-1$ or $k+1$ interfere with current symbols, causing inter-symbol interference.

Considering only frequency synchronization errors, the received signal can be expressed as

$$\mathbf{y}(k) = \mathbf{H}(k)\phi(k)\mathbf{F}(k)\mathbf{s}(k) + \mathbf{n}(k),$$

Applying receive beamforming $\mathbf{U}^{\mathrm{H}}(k)$, where

$$\mathbf{H}(k) = \mathbf{U}(k)\mathbf{\Sigma}(k)\mathbf{V}^{\mathrm{H}}(k),$$

we obtain

$$\hat{y}(k) = \underbrace{\mathbf{\Sigma}(k)\mathbf{V}^{\mathrm{H}}(k)\phi(k)\mathbf{F}(k)}_{\Phi(k)}\mathbf{s}(k) + \hat{\mathbf{n}}(k),$$

where $\hat{\mathbf{n}}(k) = \mathbf{U}^{\mathrm{H}}(k)\mathbf{n}(k)$ follows the same distribution as $\mathbf{n}(k)$, since $\mathbf{U}(k)$ is a unitary matrix. If there are no phase shifts, i.e., $\Phi(k)$ is an identity matrix, then there is no inter-layer interference. However, due to $\phi(k)$, the product $\mathbf{U}^{\mathrm{H}}(k)\phi(k)\mathbf{F}(k) \neq \mathbf{I}$ results in inter-layer interference. The interference from other streams can be calculated using the matrix

$$\Phi(k) = \mathbf{\Sigma}(k)\mathbf{V}^{\mathrm{H}}(k)\phi(k)\mathbf{V}(k).$$

In each row of the matrix, the diagonal element of $\Phi(k)$ represents the received signal stream symbol, while the off-diagonal elements indicate the inter-layer interference caused by other stream symbols. The spectral efficiency in the presence of frequency offsets is given by

$$\sum_{s=1}^{N_s} \log_2\left(1 + \frac{|\Phi_{s,s}(k)|^2}{\sum_{s'\neq s}^{N_s}|\Phi_{s,s'}(k)|^2 + \sigma^2}\right),$$

where $\Phi_{s,s}(k)$ represents the diagonal elements (signal components), $\Phi_{s,s'}(k)$ represents the off-diagonal elements (interference components), and σ^2 is the noise variance.

Similarly, the capacity degradation in the presence of both time and frequency offsets with feedback delay can be derived. The spectral efficiency can be expressed as

$$\sum_{s=1}^{N_s} \log_2\left(1 + \frac{\mathcal{A}}{\mathcal{B} + \mathcal{C} + \sigma^2}\right)$$

where $\mathcal{A} = |\Phi_{s,s}(k)|^2$ is the signal power in s-th stream, $\mathcal{B} = \sum_{s'\neq s}|\Phi_{s,s'}(k)|^2$ is inter-layer interference from the other streams due to frequency, time, and feedback delay offsets, and $\mathcal{C} = \sum_{s'}^{N_s}|\zeta_{s,s'}(k)|^2$ is the inter-symbol interference from time offsets.

IV. Simulation & Results

In this section, we present the performance plots based on the expressions derived earlier. The number of transmit and receive antennas at the transmitters and receiver is set to 2, with the number of transmitters N set to 5. The simulations were performed using MATLAB. The analysis considers that each transmitter's time and frequency offsets differ from one another. We also consider that the transmission power of each transmitter is equal and normalized so that the sum of individual powers is 0 dB. The channel elements between each transmitter and the receiver follow independent Rayleigh fading with unit variance.

TABLE I: Capacity drop for different initial phase offset ϕ values in degrees.

ϕ	SNR = 15 dB	SNR = 25 dB
3.6°	0.91%	4.19%
18°	12.0%	24.97%
36°	26.65%	42.54%
45°	33.89%	48.38%
90°	62.51%	68.93%

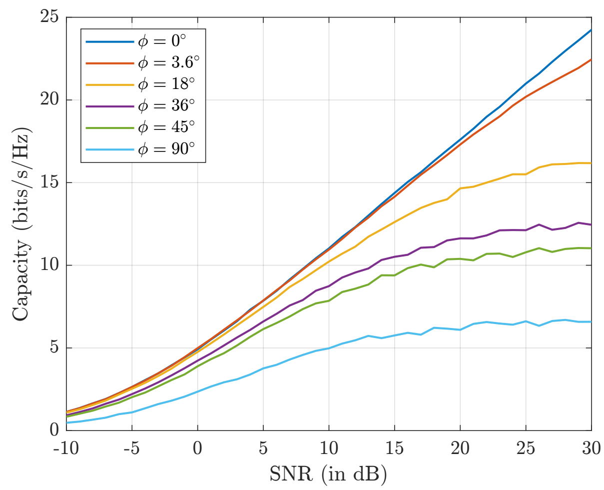


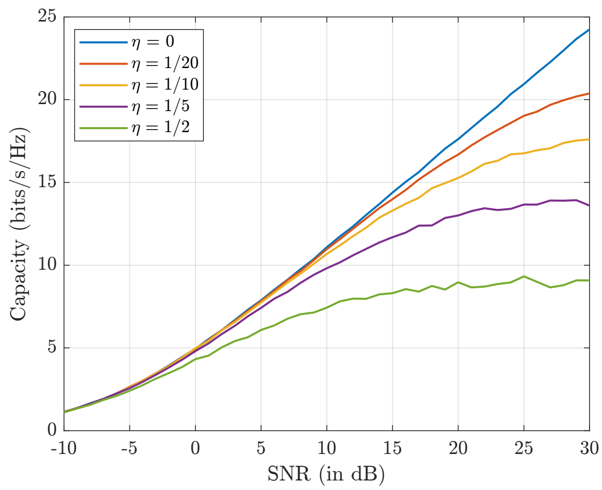
Fig. 2: Capacity versus SNR (in dB) with different initial phase offsets ϕ arising due to residual frequency offsets

Fig. 2 shows the impact of different initial phase offsets on the capacity arising due to frequency offsets f_i. The phase offsets for each transmitter are different; however, they are derived from the Gaussian distribution with the same standard deviation ϕ. The plot shows that as the initial phase offsets increase (arising due to frequency offsets), the capacity decreases. The capacity drops in percentages for SNR = 15 dB and SNR = 25 dB are tabulated in Table I.

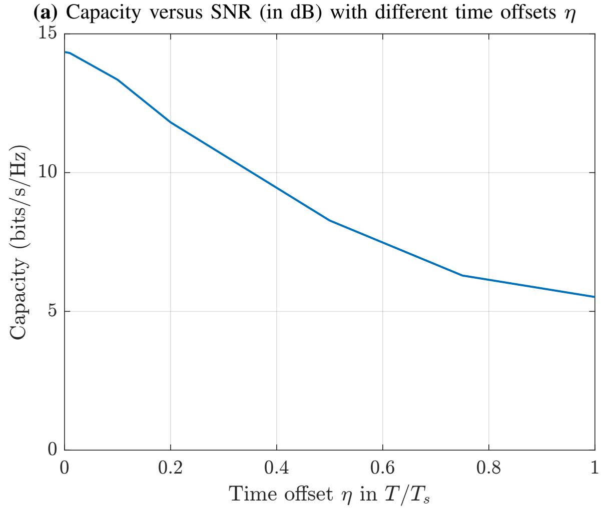
Fig. 3 illustrates the system capacity in the presence of time offsets. The time offsets η are modeled as the Gaussian distribution with various standard deviation values ranging from 0 to 1/2. Fig. 3a presents the capacity versus SNR plot, showing that at low SNRs, the impact of time offsets is minimal. Fig. 3b depicts the capacity versus time offsets for values ranging from $\eta = 0$ to one sample period ($\eta = 1$). As the time offset increases, the capacity decreases due to the loss of time synchronization. The capacity drops in percentages for SNR = 15 dB and SNR = 25 dB are tabulated in Table II.

TABLE II: Capacity drop for different time offsets η values.

η	SNR = 15 dB	SNR = 25 dB
1/20	2%	9.2%
1/10	6.9%	19.95%
1/5	18.21%	34.93%
1/2	42.20%	56.66%



(a) Capacity versus SNR (in dB) with different time offsets η



(b) Capacity versus time offsets at SNR = 15 dB

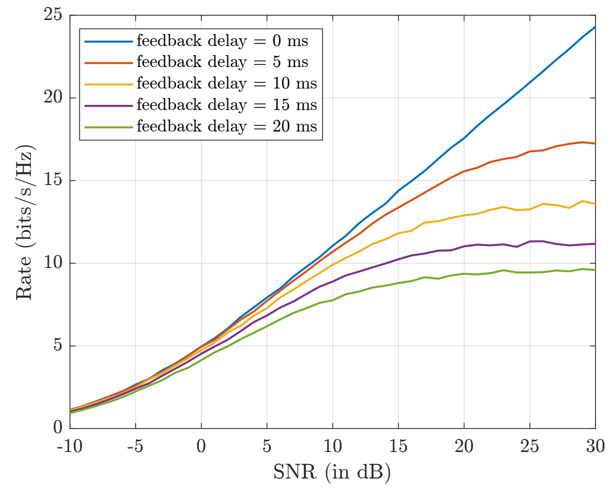
Fig. 3: Capacity analysis in the presence of time offsets η with its distribution following Gaussian distribution

Fig. 4 illustrates the system capacity in the presence of different feedback delays and Doppler spreads. Fig. 4a presents the capacity versus SNR plot, highlighting the impact of a feedback delay for a range of values from 0 ms to 20 ms at a Doppler spread of 5 Hz. Fig. 4b depicts capacity versus feedback delay for multiple Doppler frequencies. As the delay increases, the channel becomes increasingly decorrelated, leading to a significant reduction in capacity. Furthermore, it can be observed that the capacity degradation is more pronounced at higher Doppler frequencies. The capacity drops in percentages for SNR = 15 dB and SNR = 25 dB are tabulated in Table III.

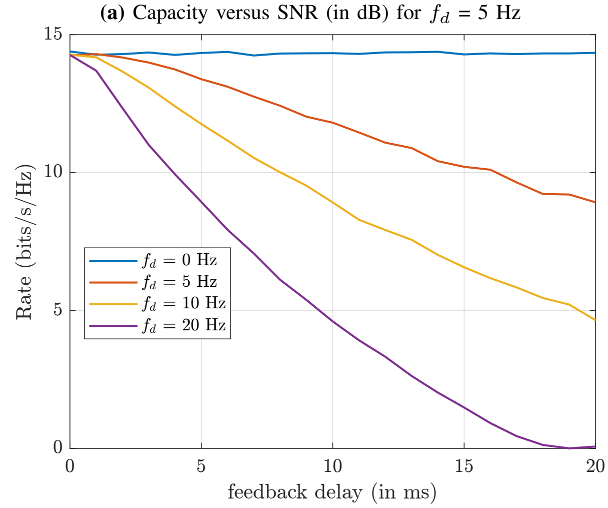
The tables and plots show that the capacity degradation due to offsets is significant under high SNR conditions. The noise impact is minimal, as it is low compared to the signal power. At high SNR, the plots show capacity performance flooring, mainly due to offsets.

TABLE III: Capacity drop for different feedback delays for f_d = 5 Hz.

τ in ms	SNR = 15 dB	SNR = 25 dB
5	4.08%	12.16%
10	10.61%	25.42%
15	19.08%	34.95%
20	26.28%	42.76%



(a) Capacity versus SNR (in dB) for f_d = 5 Hz



(b) Capacity versus channel feedback delay (in ms) with different f_d

Fig. 4: Capacity analysis in the presence of the CSI Feedback Delay

V. Conclusion

The impact of CSI feedback delay and time-frequency offsets in coherent joint transmission for distributed MIMO systems is investigated. The study focuses on transceivers with multiple antennas utilizing SVD-based precoding. The proposed analysis applies to any architecture performing coherent joint transmission. This study considered the transmit data and precoder information present at all the transmitters, not reflecting practical situations. The analysis shows that time and frequency offsets, along with CSI feedback delay, cause inter-layer interference. Additionally, time offsets result in inter-symbol interference. The amount of resources to ensure the transmit data and precoder information is present at all the transmitters should be accounted for in the capacity analysis, which reduces the overall capacity.

Future work could explore the impact of synchronization errors in non-coherent joint transmission systems and investigate the effects of synchronization on cooperating receivers. Another area of interest involves determining the number of resources required to achieve the desired level of synchronization accuracy, as it eats up the resources for coherent joint transmission, and studying the trade-off when to switch to non-coherent joint transmission if the synchronization resources needed are high for coherent joint transmission.

VI. Acknowledgement

Efforts sponsored by the U.S. Government under the Training and Readiness Accelerator II (TReX II), OTA. The U.S. Government is authorized to reproduce and distribute reprints for Governmental purposes notwithstanding any copyright notation thereon.

The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of the U.S. Government.

References

[1] International Telecommunication Union (ITU), "IMT-2030 (6G) vision and research," https://www.itu.int/en/ITU-R/study-groups/rsg5/rwp5d/imt-2030/Pages/default.aspx, 2024, accessed: December 20, 2024.

[2] X. You, C.-X. Wang, J. Huang, and et al., "Towards 6G wireless communication networks: vision, enabling technologies, and new paradigm shifts," *Sci. China Inf. Sciences*, vol. 64, no. 110301, pp. 1–22, 2021.

[3] M. Giordani, M. Polese, M. Mezzavilla, S. Rangan, and M. Zorzi, "Toward 6G networks: Use cases and technologies," *IEEE Commun. Mag.*, vol. 58, no. 3, pp. 55–61, 2020.

[4] O. Haliloglu, H. Yu, C. Madapatha, H. Guo, F. E. Kadan, A. Wolfgang, R. Puerta, P. Frenger, and T. Svensson, "Distributed MIMO systems for 6G," in *Joint European Conf. on Networks and Commun. & 6G Summit (EuCNC/6G Summit)*, 2023, pp. 156–161.

[5] K. S. Bondada, D. J. Jakubisin, K. Said, R. M. Buehrer, and L. Liu, "Wireless mobile distributed-MIMO for 6G," in *Proc. IEEE 100th Veh. Technol. Conf.*, 2024, pp. 1–7.

[6] X. Hong, Y. Jie, C.-X. Wang, J. Shi, and X. Ge, "Energy-spectral efficiency trade-off in virtual MIMO cellular systems," *IEEE J. Sel. Areas in Commun.*, vol. 31, no. 10, pp. 2128–2140, 2013.

[7] R. Heath, S. Peters, Y. Wang, and J. Zhang, "A current perspective on distributed antenna systems for the downlink of cellular systems," *IEEE Commun. Mag.*, vol. 51, no. 4, pp. 161–167, 2013.

[8] D. N. Nguyen and M. Krunz, "Cooperative MIMO in wireless networks: recent developments and challenges," *IEEE Net.*, vol. 27, no. 4, pp. 48–54, 2013.

[9] U. Madhow, D. R. Brown, S. Dasgupta, and R. Mudumbai, "Distributed massive MIMO: Algorithms, architectures and concept systems," in *Proc. Inf. Theory and Applications Workshop (ITA)*, 2014, pp. 1–7.

[10] D. Castanheira and A. Gameiro, "Distributed antenna system capacity scaling [coordinated and distributed MIMO]," *IEEE Wireless Commun.*, vol. 17, no. 3, pp. 68–75, 2010.

[11] M. Karakayali, G. Foschini, and R. Valenzuela, "Network coordination for spectrally efficient communications in cellular systems," *IEEE Wireless Commun.*, vol. 13, no. 4, pp. 56–61, 2006.

[12] W. Choi and J. G. Andrews, "Downlink performance and capacity of distributed antenna systems in a multicell environment," *IEEE Trans. Wireless Commun.*, vol. 6, no. 1, pp. 69–73, 2007.

[13] D. Gesbert, S. Hanly, H. Huang, S. Shamai Shitz, O. Simeone, and W. Yu, "Multi-cell MIMO cooperative networks: A new look at interference," *IEEE J. Sel. Areas in Commun.*, vol. 28, no. 9, pp. 1380–1408, 2010.

[14] R. Irmer, H. Droste, P. Marsch, M. Grieger, G. Fettweis, S. Brueck, H.-P. Mayer, L. Thiele, and V. Jungnickel, "Coordinated multipoint: Concepts, performance, and field trial results," *IEEE Commun. Mag.*, vol. 49, no. 2, pp. 102–111, 2011.

[15] H. Q. Ngo, A. Ashikhmin, H. Yang, E. G. Larsson, and T. L. Marzetta, "Cell-free massive MIMO: Uniformly great service for everyone," in *Proc. IEEE 16th Int. Workshop on Signal Processing Advances in Wireless Commun. (SPAWC)*, 2015, pp. 201–205.

[16] ——, "Cell-free massive MIMO versus small cells," *IEEE Trans. Wireless Commun.*, vol. 16, no. 3, pp. 1834–1850, 2017.

[17] A. Kumar, K. Appaiah, and S. R. B. Pillai, "On the ergodic sum capacity of multi-user mimo with distributed transmitter," in *proc. IEEE Int. Conf. on Commun.*, 2023, pp. 2449–2454.

[18] D. Scherber, P. Bidigare, R. ODonnell, M. Rebholz, M. Oyarzun, C. Obranovich, W. Kulp, D. Chang, and D. R. B. III, "Coherent distributed techniques for tactical radio networks: Enabling long range communications with reduced size, weight, power and cost," in *Proc. IEEE MILCOM Conf.*, 2013, pp. 655–660.

[19] A. Ozgur, O. Leveque, and D. Tse, "Spatial degrees of freedom of large distributed MIMO systems and wireless ad hoc networks," *IEEE J. Sel. Areas in Commun.*, vol. 31, no. 2, pp. 202–214, 2013.

[20] U. Demirhan and A. Alkhateeb, "Enabling cell-free massive MIMO systems with wireless millimeter wave fronthaul," *IEEE Trans. Wireless Commun.*, vol. 21, no. 11, pp. 9482–9496, 2022.

[21] S. Schwarz and S. Pratschner, "Dynamic distributed antenna systems with wireless mmwave fronthaul," in *Proc. 53rd Asilomar Conf. on Signals, Syst., and Comput.*, 2019, pp. 569–575.

[22] S. Elhoushy, M. Ibrahim, and W. Hamouda, "Downlink performance of cf massive MIMO under wireless-based fronthaul network," *IEEE Trans. on Commun.*, vol. 71, no. 5, pp. 2632–2653, 2023.

[23] J. A. Nanzer, S. R. Mghabghab, S. M. Ellison, and A. Schlegel, "Distributed phased arrays: Challenges and recent advances," *IEEE Trans. Microw. Theory Tech.*, vol. 69, no. 11, pp. 4893–4907, 2021.

[24] S. Jagannathan, H. Aghajan, and A. Goldsmith, "The effect of time synchronization errors on the performance of cooperative miso systems," in *Proc. IEEE GlobeCom Workshops.*, 2004, pp. 102–107.

[25] T.-D. Nguyen, O. Berder, and O. Sentieys, "Impact of transmission synchronization error and cooperative reception techniques on the performance of cooperative MIMO systems," in *2008 IEEE Int. Conf. on Commun.*, 2008, pp. 4601–4605.

[26] B. Zafar, S. Gherekhloo, F. Roemer, and M. Haardt, "Impact of synchronization errors on alamouti-stbc-based cooperative MIMO schemes," in *Proc. IEEE 7th Sensor Array and Multichannel Signal Processing Workshop (SAM)*, 2012, pp. 81–84.

[27] R. Mudumbai, G. Barriac, and U. Madhow, "On the feasibility of distributed beamforming in wireless networks," *IEEE Trans. Wireless Commun.*, vol. 6, no. 5, pp. 1754–1763, 2007.

[28] H. Zhou, C. Nicholls, T. Kunz, and H. M. Schwartz, "Frequency accuracy & stability dependencies of crystal oscillators," 2008. [Online]. Available: https://api.semanticscholar.org/CorpusID:58408635

[29] M. Stojanovic, J. Proakis, and J. Catipovic, "Analysis of the impact of channel estimation errors on the performance of a decision-feedback equalizer in fading multipath channels," *IEEE Trans. on Commun.*, vol. 43, no. 2/3/4, pp. 877–886, 1995.

[30] Y. Zhao, M. Zhao, L. Xiao, and J. Wang, "Capacity of time-varying rayleigh fading MIMO channels," in *Proc. IEEE 16th Int. Symposium on Pers., Indoor & Mobile Radio Commun.*, vol. 1, 2005, pp. 547–551.

[31] Herb Susmann, "Autoregressive processes are Gaussian processes," https://herbsusmann.com/2019/08/09/autoregressive-processes-are-gaussian-processes/, 2024, accessed: December 20, 2024.